



OPEN Factors influencing the intention to use telemedicine services among older adults in China

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The rapid growth of the older adult population has brought significant challenges to the medical industry. Problems such as uneven distribution of medical resources, low medical efficiency, and lack of post-hospital care have become prominent. Many hospitals in China use online operating systems and the existing telemedicine service system does not consider the needs, preferences, and characteristics of older adults. Therefore, analyzing the factors that affect older adults' intention to use telemedicine is essential for designing telemedicine service systems. The technology acceptance model (TAM) and emotional design theory were used to build a model of factors influencing older adults' telemedicine use intention in China. This research gathers data via questionnaires and employs structural equation modelling for analysis. A total of 377 participants provided responses regarding telemedicine. The collected data was subsequently analyzed using structural equation modelling. Perceived usefulness and perceived ease of use are the primary drivers affecting the behavioral intention of older adults to use telemedicine, followed by cost value, system quality, trust, and self-efficacy. In contrast, Technological anxiety has an adverse effect on the older adults' intention to use telemedicine. These findings contribute to a validated model that plays a central role in explaining telemedicine adoption among older adults. The model provides actionable insights for policymakers, healthcare providers, and designers to support the development of inclusive service strategies, improve system accessibility, and enhance user-centered design. It also offers a theoretical foundation for future research on sustained digital healthcare engagement in aging populations.

Keywords Telemedicine, Older adults, Technology acceptance model, Emotional design theory, Structural equation modeling

The rapid aging of the population and increasing life expectancy have created a huge market for healthcare services. Predictions indicate that by 2100, individuals aged 65 and older will make up 23% of the global population¹. Based on the latest comprehensive survey from China (known as the Seventh National Population Census), the share of residents aged 60 years or above has escalated to 18.70%, indicating a marked acceleration in the country's aging demography². The growth of the aging population is accompanied by an increase in age-related health problems. The physical, psychological, and social adaptations of older adults are weaker than those of younger people³, which easily affect their health.

Telemedicine has developed rapidly, especially during the COVID-19 epidemic⁴, offering a transformative solution for healthcare services by bridging geographical barriers and enhancing patient care. Telemedicine can be a complementary solution to traditional healthcare services as we can use the professional knowledge and resources of medical institutions in large cities to deliver healthcare to remote areas⁵, providing residents with convenient and cost-effective medical services through applications or related devices, alleviating medical problems caused by the lack of transportation⁶, and improving accessibility and quality of healthcare services⁷. Telemedicine facilitates reliable doctor–patient communication across time and distance, while also enhancing treatment adherence, supporting continuous monitoring, and strengthening the therapeutic relationship^{7,8}. The perceived advantages of telemedicine include improved time efficiency, reduced need for travel, and enhanced access to follow-up care and ancillary medical histories through video consultations⁹. Moreover, it plays a significant role in early diagnosis and treatment of eye diseases¹⁰, reduces the burden on emergency departments for elderly patients, and contributes to the prevention of cardiovascular diseases.

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However, telemedicine is still less popular among older adults due to several factors such as patient education, technological considerations¹¹, physiological and literacy difficulties¹², lack of reimbursements, privacy and data security issues, limited funds and technical staff, licensing issues⁶, lack of human interaction and accuracy of diagnoses and access to telehealth-capable devices¹³. Recent research on telemedicine primarily focuses on implementation processes and practical barriers, such as the lack of standardized workflows¹⁴ and key factors influencing the use of telepsychiatry¹⁵. To address delays, limited personalization, and data inefficiencies in current healthcare systems, an AI-driven framework has been proposed to enable real-time user feedback and enhance personalized, interactive care¹⁶. A proactive approach has also been used to identify and eliminate telemedicine barriers, optimize virtual patient interactions, and address disparities in care delivery¹². A recent review highlighted that digital health interventions primarily involve targeted communication and personal health monitoring for individuals, and telemedicine and decision support for providers, but research on caregivers remains limited in developing countries¹⁷.

Key limitations of telemedicine include incomplete physical and cognitive assessments⁹ and insufficient consideration of older adults' characteristics, abilities, and preferences¹⁸. Few studies have explored the factors influencing telemedicine use among older adults from a design perspective, and empirical research integrating these factors with design strategies remains scarce¹⁹. Therefore, the emotional design theory is employed in this study along with the technology acceptance model (TAM) to investigate the behavioral intention to use telemedicine among older adults in China. This study answers the question:

- (1) What are the factors that influence older adults' acceptance or adoption of telemedicine?
- (2) What design strategies can be derived from the analysis of influencing factors in telemedicine use among older adults?

Materials and methods

Development of theoretical framework and formulation of research hypotheses

In this study, TAM and emotional design theory were used to discern the determinants that affect the behavioral intention of older adults to use telemedicine. It aims to better explain the psychological and affective mechanisms underlying older adults' adoption of telemedicine. The TAM was proposed by Davis to understand the reception of information and technology systems²⁰. In TAM, perceived usefulness and perceived ease of use are core determinants of behavioral intention, as they represent the user's cognitive evaluation of the system's utility and usability. The emotional design theory was proposed by Donald²¹ and consists of three levels: visceral, behavioral, and reflective.

Even the TAM is widely used in the medical and healthcare fields²², there exists a research gap regarding the combination of TAM and emotional design theory in assessing the older adult population's inclination toward the use of telemedicine in China. Only a few studies have been carried out regarding the influence of both older adults' self-efficacy and platform system quality on their willingness to use telemedicine²². Most of the existing telemedicine products fail to consider older adults' cognitive abilities, motor skills, and emotional preferences¹⁹.

Emotional design theory aims to create products that arouse positive emotions in users and improve user engagement, satisfaction, and loyalty²³. The visceral layer greatly affects the telemedicine experiences of older patients. Without proper visual or interactive design³, usability becomes challenging for older adults, leading to technological anxiety and concerns regarding system quality due to voice interaction barriers and accidental touch issues¹⁹. Technology anxiety may negatively affect intention, consistent with affective computing literature, where emotional discomfort undermines user engagement²⁴. The behavioral layer involves interactions and usage, linked with perceived usefulness, ease of use, intentions, and enacted behavior. The reflective layer concerns users' self-perception and emotions, especially regarding their abilities, trust in telemedicine, and belief in task completion, which is collectively known as self-efficacy. Trust means believing that telemedicine can enhance health and quality of life while safeguarding user privacy and data²⁴, acknowledging that older adults' acceptance of digital health technologies is closely tied to their perception of safety and reliability. Cost values added based on actual market conditions and literature survey results²⁵. Cost value reflects the rational cost-benefit evaluation in older adults' decision-making process, particularly relevant in populations with limited digital literacy or economic resources. These interrelationships are grounded in well-established behavioral and emotional theories, offering a multi-dimensional lens to understand telemedicine adoption in aging populations. The theoretical framework developed in this study is depicted in Fig. 1.

Perceived usefulness (PU): Perceived usefulness originally refers to how much a person thinks using a certain system can improve their job performance²⁰ or a particular health intervention will enhance the individual's health status²⁵. In this article, perceived usefulness mainly refers to the professional utility and performance of telemedicine. Patients can benefit from telemedicine for specific medical services²⁶. Expanding accessibility and improving medical services positively influence perceived usefulness²⁵.

Perceived ease of use (PEOU): Perceived ease of use reflects the simplicity of operating new technologies and their intuitiveness and comfort for users²⁷. This study focuses on the older adults' time and effort required to master telemedicine systems or operate telemedicine services smoothly, with an emphasis on user perception. If a new technology or new software system is easy to use, it will be easier for older adults to understand and operate telemedicine services, and the telemedicine services will be more useful and easily accepted by users^{22,28}.

Trust (TRU): Trust is widely considered to be an important determinant in assessing the acceptance of eHealth services²⁵. The same applies to telemedicine, as trust has been confirmed as an important factor for users to accept or adopt telemedicine^{24,29}. Several studies highlighted that trust is the primary driver for adopting telemedicine. Trust reduces user uncertainty, boosts perceived usefulness and positive attitudes toward telemedicine, and strongly correlates with future online consultation and willingness to use mobile health services^{30,31}. In this study, trust refers to older adults' confidence in the reliability, professionalism, and security of telemedicine

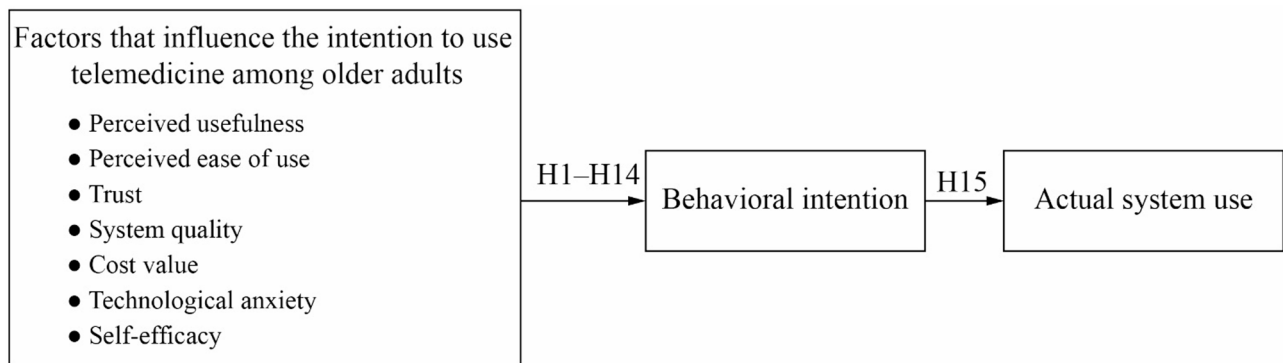


Fig. 1. Theoretical framework developed in this study.

systems and their providers, focusing on external attributes. In response to prevalent cybersecurity issues such as malware attacks, insecure download sources, unsecured Wi-Fi networks, and unencrypted communications, the deployment of encryption methods, access control mechanisms, firewalls, and malware protection is essential for securing workstation domain environments³². Effective risk management assessment is crucial for identifying vulnerabilities, evaluating risks, and implementing mitigation strategies³³. A user is unlikely to adopt technology without trust in the technology itself or its developer³⁴.

System quality (SQ): System quality refers to information quality, interaction quality, and results quality³⁵. Information quality predominantly signifies the utility of the system-provided data for patient use²², and whether the system conveys the correct medical concepts, popular science, healthcare, and nursing knowledge to users as well as prevents diseases³⁶. Interaction quality is mainly divided into three aspects³⁷. The first aspect involves assessing the logic, navigation, and design of the telemedicine system. The second aspect involves evaluating the older adults' interaction and task completion, while the third aspect involves ensuring smooth communication between doctors and patients, offering effective advice, and developing an evaluation system for online care. The results quality mainly refers to whether the services obtained by patients through online consultation are really useful. Doctor's advice and good service quality can promote the acceptance of telemedicine for older adults³⁸. Therefore, system quality is an objective evaluation of the performance and characteristics of an information system, reflecting its maturity in design and technical implementation.

Cost value (CV): Cost value positively influences users' behavioral intentions and perceived usefulness²⁵. Cost is one of the factors that may affect the integration of telemedicine^{25,39}. Cost effectiveness was identified as enabler to telemedicine use³⁹. Affordability, accessibility to healthcare, as well as time and cost savings can improve users' acceptance and satisfaction of healthcare systems²⁸, while expensive prices have been identified as a barrier to acceptance and success⁴⁰. Users can consistently track health metrics during telemedicine consultations, store treatment data in the cloud, and share information with doctors regularly⁴¹. This proactive approach facilitates timely treatment, saving time on data collection and patients' waiting for medical records, while also aiding in disease management.

Technological anxiety (TA): Technological anxiety, defined as the apprehension and panic individuals feel when contemplating or beginning to use unfamiliar information technology, centers on the user's abilities or psychological state in using technological tools and their negative reactions to computer technology²⁴, which negatively influence the acceptance or adoption of telemedicine²². Technological anxiety leads to difficulties in using health management systems⁴². Therefore, technological anxiety will affect users' inclinations to use telemedicine, and negatively affect their beliefs about using such products.

Self-efficacy (SE): Self-efficacy refers to individuals' perceptions of their capabilities and their confidence in engaging in a task⁴³, focusing on user internal capabilities. A survey has been carried out on the self-efficacy of older adults, where the older adults were requested to complete tasks without assistance or by using available resources⁴⁴. A person's beliefs can boost their self-confidence and motivate them to embrace a technology's usefulness⁴⁵. Self-efficacy is significantly related to behavioral intentions, perceived usefulness, and perceived ease of use⁴⁶.

Behavioral intention (BI): Behavioral intention mainly refers to the user's psychological tendency before a specific behavior occurs, and it is commonly considered as the proxy for acceptance of healthcare information technology⁴⁷. Behavioral intention reflects the user's inclination to utilize the system in future interactions, and the extent to which people intend to use telemedicine²⁵.

Actual system use (ASU): Actual system use refers to the actual behavior of users when using the medical service system. According to the TAM model, the user's actual usage behavior is affected by their behavioral intention⁴⁷.

Based on the preceding discussion, the variable and measurement items for the older adults' telemedicine were developed in Table 1. The research hypotheses were formulated, as tabulated in Table 2.

Considering these factors and the aforementioned dimensional variables, the structural equation model of the factors that influence the adoption of telemedicine among older adults in China was established, as shown in Fig. 2.

Constructs	Items	Measurement items	References
Perceived usefulness	PU1	Provide better medical management services	20
	PU2	Improve chances to access specialist	24
	PU3	Reduce people's psychological burden	48
	PU4	Improve the efficiency of doctor visits	26,49
Perceived ease of use	PEOU1	Learn to operate the telemedicine system quickly	20,50
	PEOU2	Interact with the telemedicine system easily	20
	PEOU3	Find it easy to use telemedicine system	20
Trust	TRU1	Improve the daily management of individual's health	24
	TRU2	Keep data and privacy confidential	34
	TRU3	Individuals on telemedicine platforms are considered trustworthy	51
System quality	SQ1	High-quality medical care and professional advices	38
	SQ2	Provide accurate and useful information on hospitals, doctors, or disease prevention	22,36
	SQ3	Maintaining the doctor-patient relationship	37
	SQ4	Integrate with offline medical service data effectively	22,36
Cost value	CV1	Save consultation time	26
	CV2	Transmit data quickly, treated effectively before the disease spreads, saving data collection time	41
	CV3	Save medical costs	40
Technological anxiety	TA1	Afraid of wrong operation and bad consequences	31,50
	TA2	Lack of emotional communication	24,50
	TA3	Could not correct operational errors	31
	TA4	Unfamiliar with telemedicine services, encounter various difficulties	52
Self-efficacy	SE1	Help solving health related problems	44
	SE2	Understand the content and function	51
	SE3	Complete medical tasks without help from others.	44
Behavioral intention	BI1	Willingness to try telemedicine system	49
	BI2	Upload medical record information	*
	BI3	Obtain relevant medical information	51
Actual system use	ASU1	Willingness to share medical experience to others	*
	ASU2	Offer advice and suggestions for the development of telemedicine services.	*
	ASU3	Tendency to use it in the consulting phase.	53
	ASU4	Recommend telemedicine to others	*
	ASU5	Participating in free telemedicine training	*

Table 1. Constructs and measurement items for older adults' telemedicine.

Factor	Abbreviation	Hypothesis
Perceived usefulness	PU	H1: Perceived usefulness has a positive effect on patients' behavioral intention to use telemedicine.
Perceived ease of use	PEOU	H2: Perceived ease of use has a positive effect on perceived usefulness of using telemedicine. H3: Perceived ease of use has a positive effect on patients' behavioral intention to use telemedicine.
Trust	TRU	H4: Trust has a positive effect on patients' perceived usefulness of using telemedicine. H5: Trust has a positive effect on patients' behavioral intention to use telemedicine.
System quality	SQ	H6: System quality has a positive effect on patients' perceived ease of use of using telemedicine. H7: System quality has a positive effect on patients' behavioral intention to use telemedicine.
Cost value	CV	H8: Cost value has a positive effect on patients' perceived usefulness of using telemedicine. H9: Cost value has a positive effect on patients' behavioral intention to use telemedicine.
Technological anxiety	TA	H10: Technological anxiety has a negative effect on patients' self-efficacy in using telemedicine. H11: Technological anxiety has a negative effect on patients' behavioral intention to use telemedicine.
Self-efficacy	SE	H12: Self-efficacy has a positive effect on patients' perceived usefulness in using telemedicine. H13: Self-efficacy has a positive effect on patients' perceived ease of using telemedicine. H14: Self-efficacy has a positive effect on patients' behavioral intention to use telemedicine.
Behavioral intention	BI	H15: Patients' behavioral intention has a positive effect on their actual system use behavior.

Table 2. Research hypotheses related to the older adults' intention to use telemedicine.

Development of questionnaire and data collection

Data were gathered through a questionnaire survey. Based on the findings of previous studies, there was no statistical significance for demographic factors on the users' behavioral intention to use telemedicine^{25,28} and thus, demographic factors were excluded from the structural equation model. As many geriatric diseases are

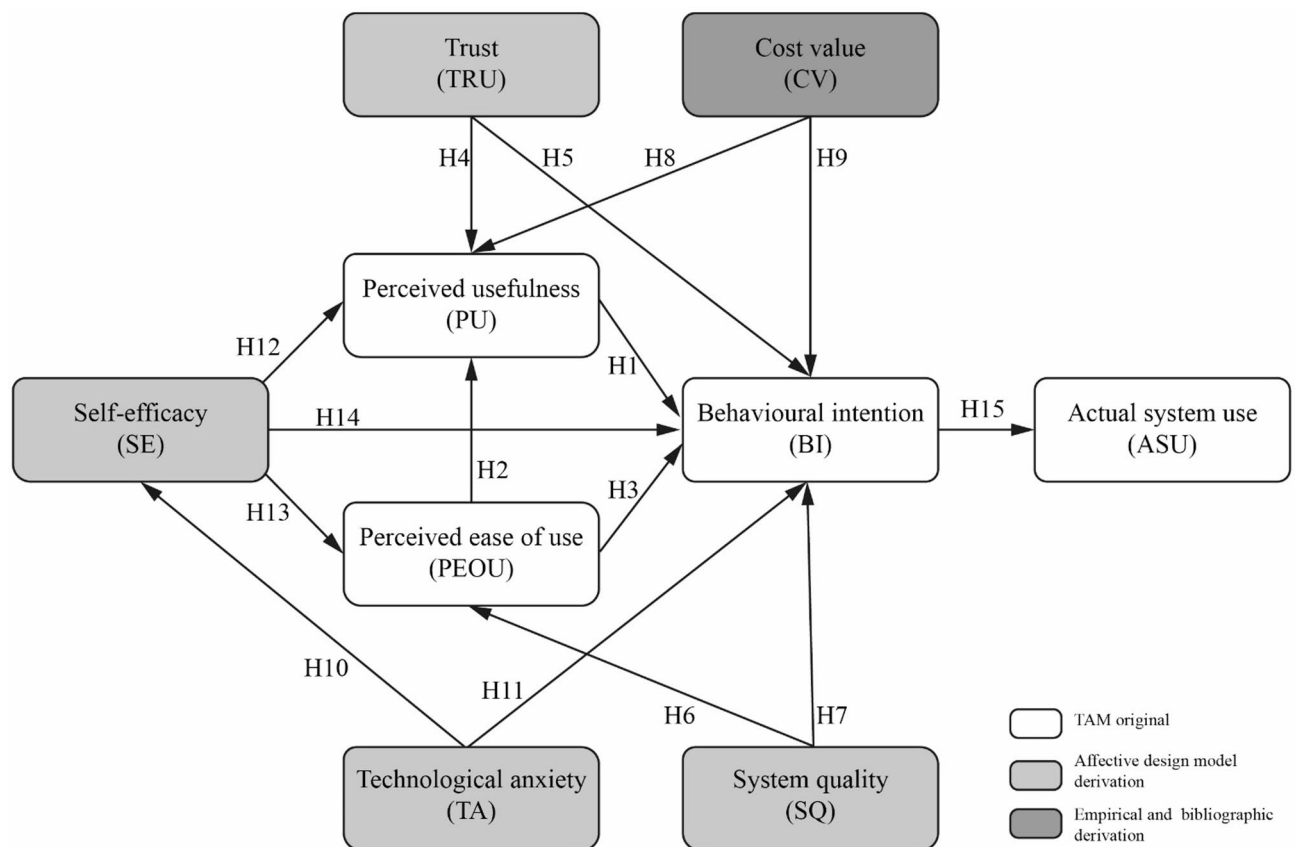


Fig. 2. Structural equation model of the factors that influence the adoption of telemedicine among older adults in China.

appearing at younger ages, people with chronic illnesses above 50 tend to start encountering health issues⁵⁴, and younger older adults tend to adopt internet technologies more quickly than their older counterparts. “old age” can be “any age older than 50,” this aimed to ensure that participants could independently engage in the study⁵⁵. The target group is defined as older adults aged 50–75 years with experience in using smartphones.

The questionnaire consists of three parts, where Part 1 is focused on gathering demographic data of the older adults, including gender, age, education level, and whether they have experience in using telemedicine. Part 2 is focused on assessing the factors that influence the older adult’s intention to use telemedicine using a five-point Likert scale. Part 3 is focused on soliciting recommendations on telemedicine. The questionnaire was modified and refined based on the findings of a pilot study as well as obtaining input from experts. This study was approved by the University of Malaya Research Ethics Committee (UM.TNC2/UMREC_2969). All participants provided written informed consent prior to their inclusion in the study. All methods were carried out in accordance with relevant guidelines and regulations.

For 377 participants, male and female respondents accounted for 52.3% and 47.7% respectively. The educational background of the respondents was concentrated in high school and below (52%), junior college (25.5%), undergraduate (15.4%), master (5.6%), PhD (1.6%). The age of the respondents is between 50 and 55 years old, accounting for 63.9%, and 17.8% of the respondents are between the ages of 56–60 years old. 6.1% and 12.2% of the respondents are between the ages of 61–65 years old and 65–75 years old respectively. Covariance-based structural equation modelling (CB-SEM) is limited by the fixed sample size whereas partial least squares structural equation modelling (PLS-SEM) can be used in models with small and complex sample sizes and is mainly used to predict key targets or identify key “driver” structures⁵⁶. Therefore, PLS-SEM was chosen to analyze the factors influencing older adults’ acceptance of telemedicine and validate the hypothesized relationships.

Results

Measurement model

Before the research hypotheses were tested, the reliability, convergent validity, and discriminant validity of the questionnaire were evaluated based on the guidelines of Hair Jr., Howard, and Ramayah et al.⁵⁶. Convergent validity was evaluated by examining the composite reliability of the criteria used (≥ 0.7), average variance extracted ($AVE \geq 0.5$), and factor loadings (≥ 0.5). This study uses Cronbach’s alpha, rho_a, and composite reliability to assess the reliability of the questionnaire⁵⁶. Table 3 shows that the composite reliability values of the PU, PEOU, TRU, SQ, CV, TA, SE, BI, and ASU constructs are all greater than 0.7, and the factor loadings and

Variable	Items	Factor loading	SMC	Mean	AVE	Cronbach's alpha	rho_a	Composite reliability
PU	PU1	0.833	0.694	3.380	0.670	0.836	0.836	0.891
	PU2	0.843	0.711					
	PU3	0.802	0.643					
	PU4	0.795	0.632					
PEOU	PEOU1	0.867	0.752	2.845	0.793	0.869	0.869	0.920
	PEOU2	0.908	0.824					
	PEOU3	0.896	0.803					
TRU	TRU1	0.845	0.714	3.054	0.717	0.805	0.817	0.884
	TRU2	0.824	0.679					
	TRU3	0.871	0.759					
SQ	SQ1	0.868	0.753	3.498	0.724	0.873	0.881	0.913
	SQ2	0.880	0.774					
	SQ3	0.831	0.691					
	SQ4	0.821	0.674					
CV	CV1	0.833	0.694	3.541	0.740	0.827	0.865	0.895
	CV2	0.899	0.808					
	CV3	0.847	0.717					
TA	TA1	0.872	0.760	3.485	0.746	0.886	0.889	0.922
	TA2	0.827	0.684					
	TA3	0.896	0.803					
	TA4	0.859	0.738					
SE	SE1	0.860	0.740	2.897	0.774	0.853	0.854	0.911
	SE2	0.862	0.743					
	SE3	0.916	0.839					
BI	BI1	0.870	0.757	3.410	0.757	0.839	0.840	0.903
	BI2	0.886	0.785					
	BI3	0.853	0.728					
ASU	ASU1	0.825	0.681	3.460	0.675	0.879	0.887	0.912
	ASU2	0.871	0.759					
	ASU3	0.809	0.654					
	ASU4	0.740	0.548					
	ASU5	0.856	0.733					

Table 3. Analysis of the reliability and validity of the questionnaire.

	PU	PEOU	TRU	SQ	CV	TA	SE	BI	ASU
PU									
PEOU	0.602								
TRU	0.589	0.610							
SQ	0.649	0.452	0.542						
CV	0.258	0.284	0.305	0.412					
TA	0.327	0.510	0.385	0.261	0.167				
SE	0.465	0.417	0.331	0.280	0.065	0.473			
BI	0.732	0.711	0.636	0.606	0.416	0.504	0.492		
ASU	0.557	0.425	0.482	0.581	0.374	0.273	0.321	0.724	

Table 4. Discriminant validity of the questionnaire according to the HTMT method.

AVE values are all greater than 0.5, indicating that the questionnaire has favorable reliability and convergent validity.

Discriminant validity was assessed using the Heterotrait-monotrait (HTMT) method proposed by Henseler⁵⁷. The closer the HTMT value between two variables is to 0.85, the worse the discriminant validity is. As can be seen from Table 4, the HTMT values among PU, PEOU, TRU, SQ, CV, SI, TA, SE, BI, and ASU are all lower than the test standard of 0.85, which shows that the questionnaire has good discriminant validity.

Structural equation model

The structural equation model was developed to identify the relationships between the various constructs within the research model⁵⁸. By observing the impact size and significance of each path in Table 5, it can be ascertained that PU ($\beta = 0.254, p < 0.001$), PEOU ($\beta = 0.238, p < 0.001$), TRU ($\beta = 0.115, p < 0.05$), SQ ($\beta = 0.127, p < 0.05$), CV ($\beta = 0.137, p < 0.001$), and SE ($\beta = 0.108, p < 0.05$) all have a significant positive effect on behavioral intention. In addition, PEOU ($\beta = 0.275, p < 0.001$), TRU ($\beta = 0.276, p < 0.001$), and SE ($\beta = 0.214, p < 0.001$) have a significant effect on perceived usefulness whereas CV ($\beta = 0.074, p > 0.05$) does not. Both SQ ($\beta = 0.329, p < 0.001$) and SE ($\beta = 0.278, p < 0.001$) have a significant positive effect on perceived ease of use. Conversely, technological anxiety negatively affects self-efficacy ($\beta = -0.413, p < 0.001$) and behavioral intention ($\beta = -0.126, p < 0.05$). Behavioral intention positively affects actual system use ($\beta = 0.625, p < 0.001$). Based on the results, all of the research hypotheses formulated in this study are supported except for H8, H10, and H11.

Discussion

This study is the first study to combine TAM and emotional design theory to evaluate the factors that influence older adults’ intention to use telemedicine in China. The results indicate that perceived usefulness, perceived ease of use, trust, system quality, self-efficacy, and cost value significantly and positively influence the behavioral intention of older adults to use telemedicine. There is a significant positive relationship between perceived ease of use and perceived usefulness, which is consistent with the findings of Kung²⁹. These findings highlight the significant impact of cognitive and emotional factors on telemedicine acceptance, which is in accordance with the study by Zobair⁵⁹. Therefore, when implementing and promoting telemedicine, besides considering its benefits to patients, factors like information security, system quality, user experience, and overall operational costs should be taken into account, this is supported by Lu⁶⁰. Especially, designers and software developers need to evaluate if the product matches the cognition and emotions of the elderly¹⁹.

Technological anxiety hinders the behavioral intention to use telemedicine among older adults, acting as a primary barrier to its adoption⁵⁰. Technological anxiety can provoke resistance and fear of new technology, potentially reducing users’ self-efficacy. Individuals possessing greater self - efficacy tend to tackle challenges effectively⁵¹. High self-efficacy leads users to perceive telemedicine as more useful and user-friendly, increasing their willingness to try it. Trust in technology also enhances its perceived usefulness by fostering positive psychological associations. It is crucial to focus on users’ capacity for innovation and self - assurance in operating information technology, free from the fear of errors, especially since a significant portion of users will not adopt a technology if they do not trust it, this is agree with AlQudah³⁴. As Niu⁶¹ suggest, trust in product safety and service providers can respectively increase the usage depth and breadth of telemedicine products. In addition, system quality significantly improves perceived ease of use. High system quality in terms of information, service, and interaction makes the system seem user-friendly, increasing users’ willingness to use it. System quality determines the user experience and ease of operation in telemedicine systems, driven by its optimization in stability, compatibility, and low cognitive loads. Governments and healthcare providers can implement incentives, such as subsidies for telemedicine services and programs to enhance digital literacy among older adults¹¹. Addressing technological anxiety and improving system quality can significantly boost telemedicine acceptance in this demographic.

These findings have multifaceted implications. Theoretically, they underscore the importance of integrating self-efficacy and trust into technology acceptance models, particularly for vulnerable groups like older adults. Practically, they highlight the need for designers and developers to prioritize enhancing the user-friendliness and reliability of telemedicine systems. These factors influencing telemedicine adoption among older adults in China offer valuable insights for creating user-centric systems that cater to their specific needs. Categorizing users and

	Path	β	Stan- dard error	t-value	p- value	2.50%	97.50%	Statistically significant ($p < 0.05$)
H1	PU→BI	0.254	0.054	4.667	0.000	0.144	0.357	Yes
H2	PEOU→PU	0.275	0.063	4.357	0.000	0.152	0.403	Yes
H3	PEOU→BI	0.238	0.051	4.704	0.000	0.128	0.329	Yes
H4	TRU→PU	0.276	0.072	3.826	0.000	0.133	0.412	Yes
H5	TRU→BI	0.115	0.050	2.286	0.022	0.012	0.207	Yes
H6	SQ→PEOU	0.329	0.059	5.599	0.000	0.207	0.436	Yes
H7	SQ→BI	0.127	0.051	2.506	0.012	0.027	0.225	Yes
H8	CV→PU	0.074	0.056	1.323	0.186	-0.037	0.180	No
H9	CV→BI	0.137	0.035	3.908	0.000	0.065	0.202	Yes
H10	TA→SE	-0.413	0.057	7.230	0.000	-0.516	-0.291	Yes
H11	TA→BI	-0.126	0.059	2.127	0.033	-0.248	-0.019	Yes
H12	SE→PU	0.214	0.050	4.296	0.000	0.117	0.315	Yes
H13	SE→PEOU	0.278	0.058	4.778	0.000	0.163	0.391	Yes
H14	SE→BI	0.108	0.047	2.321	0.020	0.018	0.200	Yes
H15	BI→ASU	0.625	0.033	18.661	0.000	0.550	0.684	Yes

Table 5. Results of hypothesis testing.

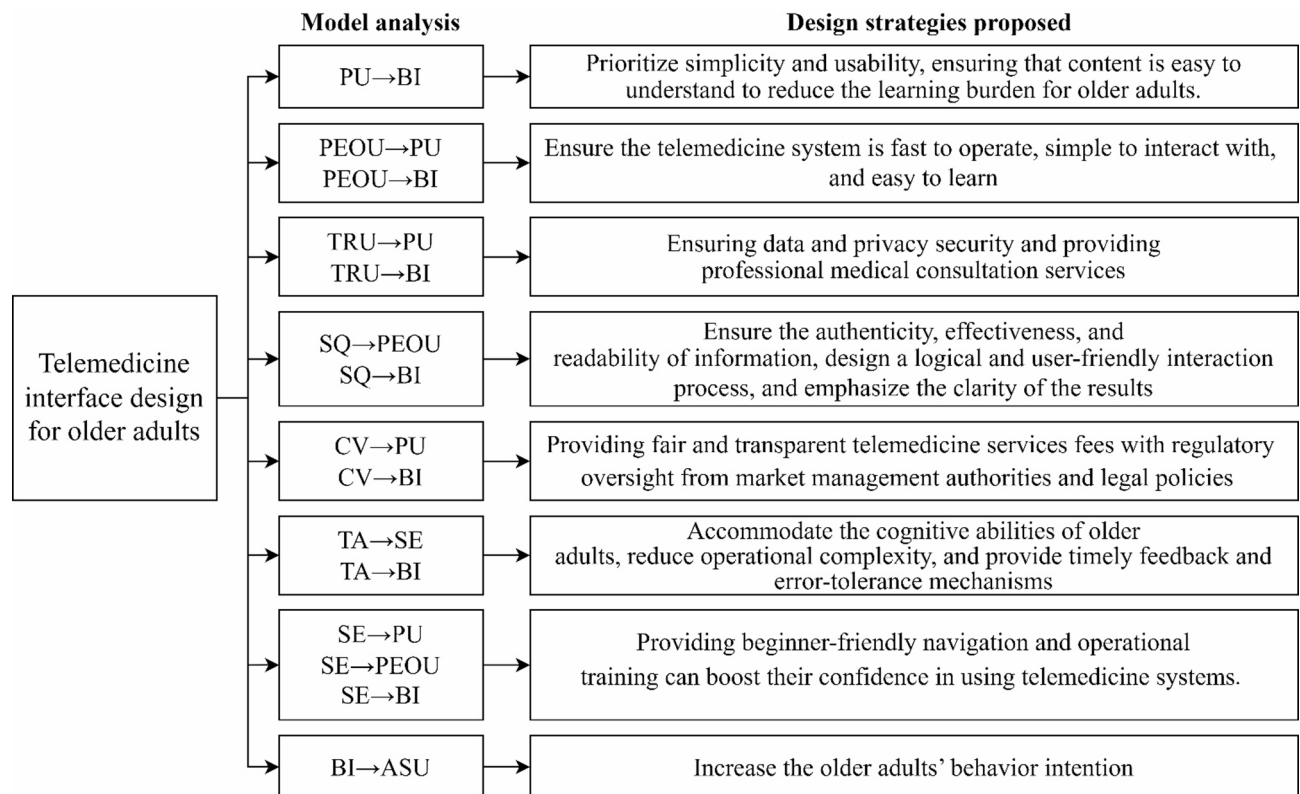


Fig. 3. Relationship between influencing factor model and design strategy.

comprehending their requirements enables the customization of services, thereby enhancing healthcare delivery, which calls for a user - centered approach in service design, including more training, support, guidance, and information⁶². To design an effective telemedicine system for older adults, first, it is essential to understand their functional and emotional needs, and then translate these needs into design specifications that highlight the system's utility. Second, it is important to prioritize features based on the needs of older adults, focusing on essential and frequently used functions, while ensuring that the graphical user interface remains simple. Third, it is crucial to ensure that the older adults' privacy, safety, and security are protected. The extensive exchange of sensitive and critical information between users and applications renders such data highly vulnerable to cyber threats, including data theft and the leakage of confidential information⁶³. To prevent data breaches and privacy violations involving doctors and patients, it is crucial to mitigate risks such as the misuse of cloud resources for fraudulent or malicious purposes due to unidentified or unauthorized access, insecure interfaces, or inadequate data handling practices⁶⁴. In addition, it is important to enhance the credentials of online doctors and implement government policies with robust oversight to foster trust in telemedicine among older adults. Fourth, it is necessary to strengthen telemedicine promotion, offer user training and public health education campaigns¹³, provide remote assistance options for the children of older adults, implement intuitive interface navigation, and add operational guidance. This can be achieved by improving the fault tolerance of the graphical user interface and alleviate the older adults' concerns to boost their confidence and reduce technological anxiety. Fifth, it is essential to develop older adult-friendly designs and personalized recommendations. Moreover, there is a need to make the prices of telemedicine services transparent, with collaborative regulation by the government, hospitals, and businesses. It is proposed that the prices of telemedicine services should be at par with or lower than in-person services and online medical services should be covered by medical insurance to encourage telemedicine adoption and reduce the burden on traditional healthcare systems. Based on the SEM model analysis, the design strategy proposed in Fig. 3.

However, this research possesses certain constraints that deserve recognition. First, the sample size of 377 participants, while adequate for structural equation modeling, is predominantly drawn from Guangdong, China, with limited regional diversity. This restricts the generalizability of the findings to the broader population of older adults in China and necessitates caution when applying the results to other countries. Second, A longitudinal study would provide stronger evidence on factors influencing telemedicine adoption over time. Finally, while the questionnaire was developed based on previous studies and refined through a pilot test, details on its content and face validity were not provided. Future research should ensure the questionnaire accurately measures the constructs of interest and is suitable for the target population, with validation steps clearly outlined.

Conclusion

This study makes a novel contribution by integrating the Technology Acceptance Model (TAM) with affective design principles to examine the factors influencing older adults' intention to use telemedicine services in China. By extending the traditional TAM framework to include emotional and user-centered variables such as trust, self-efficacy, technology anxiety, and system quality, the study offers a more comprehensive understanding of telemedicine adoption. The proposed model highlights both cognitive and affective determinants, emphasizing that emotional factors play a critical role alongside perceived usefulness and ease of use in shaping user intentions.

Moreover, the study identifies key barriers preventing telemedicine use among older adults, including distrust, perceived complexity, technological anxiety, and learning difficulties. By addressing these issues, it provides actionable design strategies to enhance system usability, emotional comfort, and accessibility. This study aids technology developers in gaining a deeper understanding of the elderly's behavior patterns, encourage medical practitioners to pay attention to telemedicine systems, and improve the quality, ease of use, and usability of their systems. These findings provide reference value for telemedicine technology developers, system designers, medical personnel and governments, and provide a direction for promoting the use of telemedicine systems by older adults.

Based on these findings, design specifications are proposed for designers and developers. First, it is crucial to enhance the utility and ease of use of telemedicine for older adults, reduce learning time and cognitive barriers, and ensure that telemedicine services are covered by medical insurance. In addition, it is important to boost service quality in terms of information and interaction, and elevate medical and expert credentialing to build trust among older adults. It is recommended that technology support, privacy security, and individual service should be provided to encourage older adults to use telemedicine and increase their self-efficacy. Trainings should be offered to improve system resilience and alleviate technological anxiety. In the future, it is also necessary to enhance the adaptability of telemedicine service scenarios, establish a collaborative network between communities and families for support, promote telemedicine through multiple channels, and strengthen policy support and social security. Future studies can be carried out to analyze and compare telemedicine usage and the factors that influence the intention to use telemedicine between older adults in China and Western countries, providing insights into the global telemedicine market for older adults. Future research also can explore the dynamic and longitudinal changes in older adults' adoption of telemedicine, including shifts in attitudes and behaviors over time. Expanding the study to diverse regions and socio-economic groups across China would enhance the generalizability of the proposed model. Integrating additional theoretical frameworks, such as the Health Belief Model (HBM), may offer deeper insights into the psychological, cultural, and contextual factors influencing telemedicine use. Interdisciplinary collaboration with designers, engineers, and healthcare professionals is also encouraged to support the development of user-centered telemedicine interfaces that are both emotionally responsive and technologically accessible.

Data availability

The datasets generated and/or analysed during the current study are not publicly available due to the policy of privacy protection but are available from the corresponding author on reasonable request.

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H.H.Q: Conceptualization, Methodology, Writing – Original Draft, Data Curation, Formal Analysis, Visualization. S.H.A.R: Supervision, Conceptualization, Writing – Reviewing & Editing. R.A.R.G.: Supervision, Methodology, Writing – Review & Editing. All authors interpreted the results, and revised and approved the submission of the final article.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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